

MATD10: Project 3 Report

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Introduction

In this project, we are still focusing on the advection equation with the same initial conditions as in Project 2:

$$\begin{cases} u_t + cu_x = 0, & a \leq x \leq b, 0 \leq t \leq T, \\ u(x, 0) = f(x), & a \leq x \leq b, \\ u(a, t) = g(t), & 0 \leq t \leq T. \end{cases}$$

We use different SBP-SAT schemes to solve this IBVP and compare their properties to the schemes with ghost node construction in Project 2. The exact solution that we will employ is

$$u(x, t) = \exp(-(x - ct)^2) + \cos(13(x - ct))$$

with parameters $a = -1$, $b = 1$, $c = 1.1$, and $T = 0.5$. The grid sizes will be the same as in Project 2 and we will vary T to understand how the schemes behave with an increase in final time: $nx = 10, 25, 50, 100, 200, 600, 1000$, $T = 0.5, 1.0, 1.5, 2.0$.

Results

Identity Verification

```
n = 10, dx = 0.2222222222222222
SBP21: error = 0.000e+00
SBP42: error = 3.539e-16
SBP63: error = 6.279e-01
SBP44: error = 1.834e-15

n = 25, dx = 0.08333333333333333
SBP21: error = 0.000e+00
SBP42: error = 2.394e-16
SBP63: error = 3.990e-16
SBP44: error = 1.073e-15

n = 50, dx = 0.04081632653061224
SBP21: error = 0.000e+00
SBP42: error = 2.498e-16
SBP63: error = 4.966e-16
SBP44: error = 1.949e-15

n = 100, dx = 0.020202020202020204
SBP21: error = 1.110e-16
SBP42: error = 2.359e-16
SBP63: error = 4.406e-16
SBP44: error = 1.351e-15

n = 200, dx = 0.010050251256281407
SBP21: error = 0.000e+00
SBP42: error = 3.469e-16
SBP63: error = 4.137e-16
SBP44: error = 1.454e-15

n = 600, dx = 0.00333889816360601
SBP21: error = 0.000e+00
SBP42: error = 2.359e-16
SBP63: error = 6.353e-16
SBP44: error = 1.576e-15

n = 1000, dx = 0.002002002002002002
SBP21: error = 0.000e+00
SBP42: error = 3.400e-16
SBP63: error = 3.968e-16
SBP44: error = 1.280e-15
```

Figure 1: Verification of the identity condition for the SBP-SAT scheme

As can be seen from the figure above, all SBP schemes satisfy the identity condition, as the uniform norm of the difference is close to machine precision for almost all grid sizes. It is worth noting that for SBP63 when $n = 10$, the error is large compared to the remaining results, this is due to how the matrices are constructed for SBP63 instead of a failure of the construction itself. The SBP63 operator consists of two wide boundary blocks together with an interior scheme, where the boundary blocks are $m \times m$ matrices. Hence, when the number of grid points is too small (i.e. $n < 2m$), the two boundary blocks would overlap and this is poorly defined, and it leads to the observation of large errors of SBP63 for small grid sizes.

Additionally, we note that SBP21 has an error of zero for all grid sizes. This is not an indication that SBP21 is the best among all schemes, but rather a consequence of how the SBP21 operator is constructed. SBP21 has the simplest stencil and also the diagonal H with clean entries compared to the others. Therefore, the cancellations we have in $HD + D^T H - B$ are much more exact in SBP21 with float arithmetic, compared to the more complex SBP42 and SBP63, where roundoff errors are more unavoidable due to its complicated entries.

Observed Order of D

SBP21 (formal order $p = 1$)		SBP63 (formal order $p = 3$)	
k	error	k	error
0	0.000e+00	0	7.048e-13
1	7.105e-15	1	7.252e-13
2	2.020e-02	2	7.407e-13
3	6.020e-02	3	7.625e-13
4	1.196e-01	4	1.026e-04
5	1.980e-01	5	4.959e-04
6	2.950e-01	6	1.438e-03
7	4.102e-01	7	3.246e-03
number of exact polynomial tests passed = 2		number of exact polynomial tests passed = 4	
highest exact degree = 1		highest exact degree = 3	
SBP42 (formal order $p = 2$)		SBP44 (formal order $p = 4$)	
k	error	k	error
0	1.776e-15	0	1.554e-14
1	1.421e-14	1	1.510e-14
2	1.465e-14	2	1.599e-14
3	1.032e-03	3	2.043e-14
4	4.054e-03	4	4.947e-05
5	9.949e-03	5	2.414e-04
6	1.954e-02	6	7.066e-04
7	3.358e-02	7	1.609e-03
number of exact polynomial tests passed = 3		number of exact polynomial tests passed = 4	
highest exact degree = 2		highest exact degree = 3	

Figure 2: Order of D for the SBP-SAT scheme

The observed order of D aligns well with the theoretical order for SBP21, SBP42, and SBP63. Recall that the order of D is p as defined in

$$D(x^k) = kx^{k-1} \quad \text{for } k = 0, \dots, p$$

Our results suggest that the numbers of exact polynomial tests passed are 2, 3, 4, 4 with the order of D i.e. the highest exact polynomial degree being 1, 2, 3, 3.

Convergence Test

SBP21						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.125	1.1027566	-	0.91920168	-
25	0.08333333	0.05555556	1.4075982	-0.249	1.1843108	-0.258
50	0.040816327	0.029411765	0.32900641	2.036	0.30967756	1.879
100	0.02020202	0.014285714	0.093695914	1.786	0.076213491	1.993
200	0.010050251	0.0072463768	0.023577556	1.976	0.018863716	2.000
600	0.0033388982	0.0024271845	0.0026439028	1.986	0.0020818534	2.000
1000	0.002002002	0.0014534884	0.00094756263	2.006	0.00074846031	2.000
SBP42						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.125	1.0941161	-	1.00365	-
25	0.08333333	0.05555556	0.42930608	0.954	0.3199619	1.166
50	0.040816327	0.029411765	0.04846671	3.056	0.028498186	3.388
100	0.02020202	0.014285714	0.0051250728	3.195	0.0028724675	3.263
200	0.010050251	0.0072463768	0.00058025496	3.120	0.00032825838	3.107
600	0.0033388982	0.0024271845	2.2095941e-05	2.966	1.1655533e-05	3.029
1000	0.002002002	0.0014534884	4.7931101e-06	2.988	2.501887e-06	3.008
SBP63						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.125	1.22895	-	1.1998062	-
25	0.08333333	0.05555556	0.3895854	1.171	0.28405524	1.469
50	0.040816327	0.029411765	0.040887104	3.158	0.025165541	3.396
100	0.02020202	0.014285714	0.0036195653	3.447	0.0016869738	3.843
200	0.010050251	0.0072463768	0.00051555373	2.791	0.00011775505	3.813
600	0.0033388982	0.0024271845	1.0163467e-05	3.563	1.5577652e-06	3.925
1000	0.002002002	0.0014534884	1.1811402e-06	4.288	1.9909265e-07	4.022
SBP44						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.125	1.2395991	-	1.0321422	-
25	0.08333333	0.05555556	0.38712032	1.187	0.2945847	1.278
50	0.040816327	0.029411765	0.029415852	3.611	0.018869675	3.850
100	0.02020202	0.014285714	0.0018107874	3.964	0.0011621004	3.963
200	0.010050251	0.0072463768	0.00011716087	3.922	7.1704098e-05	3.990
600	0.0033388982	0.0024271845	1.3946315e-06	4.021	8.7572409e-07	3.998
1000	0.002002002	0.0014534884	1.7994671e-07	4.003	1.1322328e-07	3.999

Figure 3: Convergence test results for the SBP-SAT scheme

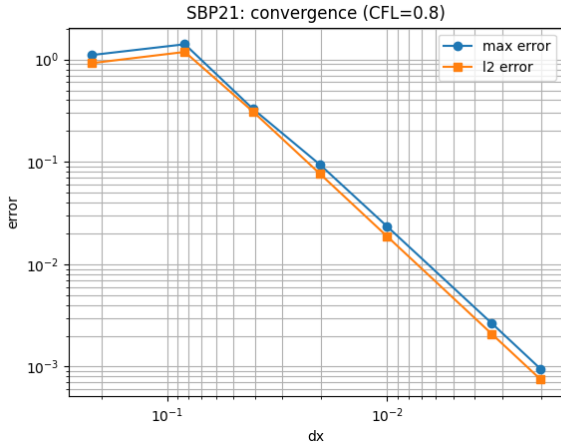
For both norms, we conclude that their observed orders of convergence are 2, 3, 4, 4 for SBP21, SBP42, SBP63, and SBP44 respectively. First observation we make is that by looking at max_err and l2_err, we see that l2_err is smaller for all schemes and all grid sizes, which is expected because it averages out the errors across the grid, while max_err focuses on the largest error at any point.

Additionally, for SBP($2p, p$) schemes, the order of convergence is one degree higher than the expected order. The schemes were benefited from the symmetrical structure of the diagonal matrix H with the associated boundary stencils, which produced one more degree of accuracy compared with the boundary accuracy alone. For SBP44, the order of convergence is the same as the expected order, which is due to the fact that the boundary accuracy is the same as the interior accuracy, for which our result aligns.

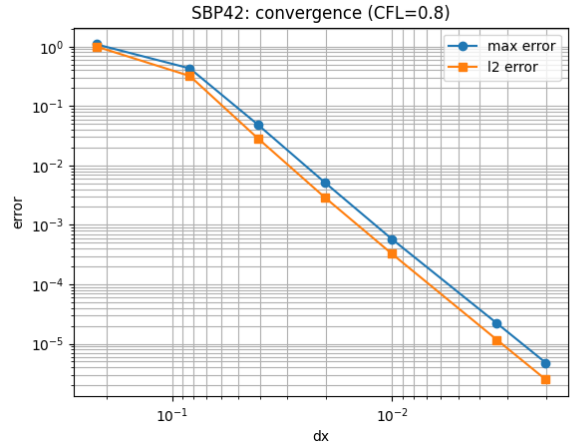
By testing the order of convergence with finer grid sizes, we can see that the uniform norm order of convergence for SBP63 and SBP44 begins to deteriorate on these finer grids. It is unlikely that these results indicate instability, because the l2_err and p_l2 still align well, and the maximum error continues to decrease rather than blow up. The more plausible explanation might be that since the error is already very small which is close to machine error, other systemic errors such as roundoff errors become more dominating and thus the observed order of convergence is not as expected.

SBP21						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
100	0.02020202	0.014285714	0.093695914	-	0.076213491	-
4000	0.00050012503	0.00036363636	5.9167198e-05	1.992	4.6708315e-05	2.000
7500	0.00026670223	0.0001939488	1.7003466e-05	1.983	1.3282813e-05	2.000
10000	0.00020002	0.00014543339	9.591973e-06	1.990	7.471083e-06	2.000
SBP42						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
100	0.02020202	0.014285714	0.0051250728	-	0.0028724675	-
4000	0.00050012503	0.00036363636	7.5178905e-08	3.009	3.8854256e-08	3.031
7500	0.00026670223	0.0001939488	1.1404285e-08	3.000	5.8892815e-09	3.001
10000	0.00020002	0.00014543339	4.809712e-09	3.001	2.4839432e-09	3.000
SBP63						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
100	0.02020202	0.014285714	0.0036195653	-	0.0016869738	-
4000	0.00050012503	0.00036363636	3.113919e-09	3.776	7.7343423e-10	3.946
7500	0.00026670223	0.0001939488	2.8127717e-10	3.824	6.2709192e-11	3.996
10000	0.00020002	0.00014543339	1.0654461e-10	3.374	1.9870971e-11	3.994
SBP44						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
100	0.02020202	0.014285714	0.0018107874	-	0.0011621004	-
4000	0.00050012503	0.00036363636	7.116443e-10	3.988	4.4174353e-10	3.997
7500	0.00026670223	0.0001939488	2.1874075e-10	1.876	3.6056149e-11	3.985
10000	0.00020002	0.00014543339	1.2418233e-10	1.968	1.1577347e-11	3.948

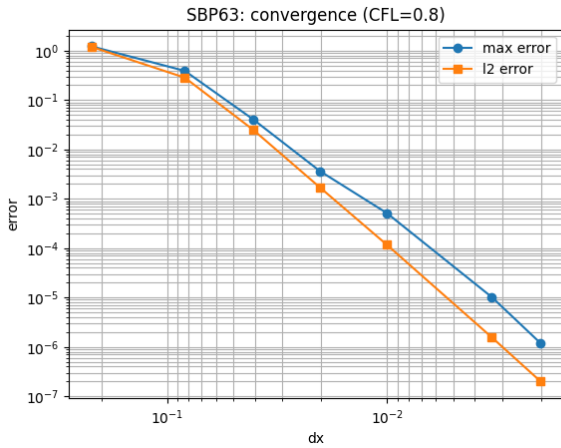
Figure 4: Further Convergence Test Results



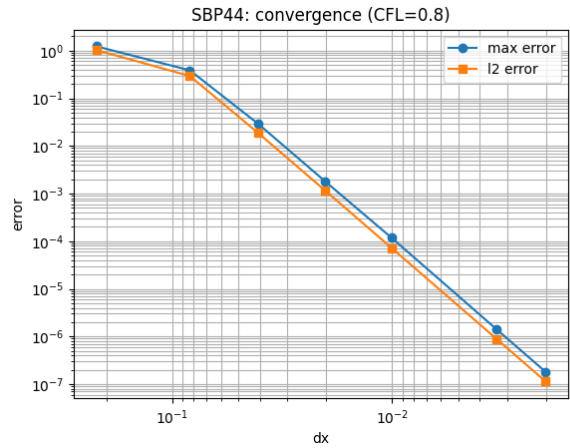
(a) SBP21 convergence plot



(b) SBP42 convergence plot



(c) SBP63 convergence plot



(d) SBP44 convergence plot

Now we present whether convergence behavior varies with different **final time** T . We first see that the observed order of convergence is still roughly the same and does not significantly change with different T . However, what did change here is the size of the error. For a fixed grid size, when T is larger, both the `max_err` and `l2_err` are larger. This is expected, since the error would accumulate over time, and thus a larger T would lead to a larger error.

A second observation is that the error for each of the schemes fluctuates more with coarser grid sizes when T is larger. Especially for SBP63 and SBP44. In particular, even when `max_err` is more sensitive on coarse grids or near the boundary, `l2_err` remains more stable and continues to show the expected convergence behavior. However, once the grid is fine enough, the expected orders of convergence are approximately 2, 3, 4, and 4 again, which aligns with the results we have for $T = 0.5$.

Lastly, we compare the results of Project 3 with the results of Project 2. A key point to take away is that both the ghost node construction and the SBP-SAT scheme use a lower order of accuracy at the boundaries, but the distinction is that the SBP-SAT scheme achieves a higher global convergence order, while the ghost node construction has a lower global convergence order that was more negatively affected by the lower order at the boundary. The boundary affects both, but more significantly for the latter. Hence, the construction of SBP-SAT schemes is more sophisticated and treats the boundary in a more robust way, while ghost nodes and one-sided difference formulas can reduce the overall accuracy more directly with the solution being more severely affected by the boundary errors, even if they might be easier to implement.

Another difference between Project 2 and Project 3 is the advancement method. Project 2 used RK4, while Project 3 uses RK6 to evolve the semi-discrete SBP-SAT system. This higher-order time stepping helps reduce time truncation error, especially on fine grids, but it also increases the computational cost. Therefore, some of the improved accuracy in Project 3 comes not only from the SBP-SAT boundary treatment, but also from the use of a more accurate and more expensive advancement method.

To conclude, the improvement comes with a tradeoff in computational cost and efficiency. The SBP-SAT schemes require carefully constructing matrix operators with more complex stencils once the order of the scheme increases which requires more sophisticated implementation. Similarly with SBP44, where a non-diagonal matrix is used, which would significantly increase the computational cost. Therefore, Project 3 offers better stability and convergence behavior, while Project 2 remains simpler and cheaper computationally.

SBP21						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.14285714	0.87919167	-	0.89209443	-
25	0.08333333	0.058823529	2.038023	-0.857	1.779245	-0.704
50	0.040816327	0.029411765	0.73335456	1.432	0.55847212	1.623
100	0.02020202	0.014492754	0.17639497	2.026	0.13525768	2.016
200	0.010050251	0.0072992701	0.043535038	2.004	0.033167979	2.013
600	0.0033388982	0.0024271845	0.0047665885	2.007	0.003644884	2.004
1000	0.002002002	0.0014556041	0.0017154857	1.998	0.0013095949	2.001
SBP42						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.14285714	1.1998539	-	1.031193	-
25	0.08333333	0.058823529	0.75564044	0.471	0.56887329	0.606
50	0.040816327	0.029411765	0.068105158	3.372	0.045933491	3.526
100	0.02020202	0.014492754	0.0061408679	3.421	0.0043215767	3.361
200	0.010050251	0.0072992701	0.00062378904	3.276	0.00048252299	3.140
600	0.0033388982	0.0024271845	2.4612553e-05	2.933	1.7030817e-05	3.035
1000	0.002002002	0.0014556041	5.4514689e-06	2.947	3.6553577e-06	3.009
SBP63						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.14285714	1.3860945	-	1.2211155	-
25	0.08333333	0.058823529	0.48156392	1.078	0.43178198	1.060
50	0.040816327	0.029411765	0.038850291	3.527	0.030850788	3.697
100	0.02020202	0.014492754	0.0043675464	3.108	0.0021645429	3.778
200	0.010050251	0.0072992701	0.0004980769	3.110	0.00014743482	3.848
600	0.0033388982	0.0024271845	2.9721007e-06	4.648	1.8307164e-06	3.983
1000	0.002002002	0.0014556041	4.5158964e-07	3.684	2.4427838e-07	3.938
SBP44						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.14285714	1.5693713	-	1.1278039	-
25	0.08333333	0.058823529	0.67668176	0.858	0.51699899	0.795
50	0.040816327	0.029411765	0.047377032	3.725	0.03202501	3.897
100	0.02020202	0.014492754	0.0029576512	3.944	0.0019611505	3.971
200	0.010050251	0.0072992701	0.00017669844	4.036	0.00012079168	3.992
600	0.0033388982	0.0024271845	2.1486467e-06	4.002	1.473218e-06	3.999
1000	0.002002002	0.0014556041	2.7872749e-07	3.993	1.9041736e-07	4.000

Figure 6: Convergence test results for the SBP-SAT scheme with $T = 1.0$

SBP21						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.15	0.71065752	-	0.66367552	-
25	0.083333333	0.06	2.1687044	-1.138	1.7076088	-0.964
50	0.040816327	0.029411765	0.92061369	1.200	0.72321401	1.204
100	0.02020202	0.014563107	0.25797437	1.809	0.16916106	2.066
200	0.010050251	0.0072815534	0.064663712	1.982	0.040963566	2.031
600	0.0033388982	0.0024271845	0.0071662081	1.996	0.0044830307	2.008
1000	0.002002002	0.0014548982	0.0025765704	2.000	0.0016100893	2.002
SBP42						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.15	1.1566847	-	0.93910652	-
25	0.083333333	0.06	1.0419252	0.107	0.75084607	0.228
50	0.040816327	0.029411765	0.082364607	3.555	0.054791357	3.667
100	0.02020202	0.014563107	0.0069472929	3.516	0.0047643561	3.473
200	0.010050251	0.0072815534	0.00066672144	3.357	0.00051779101	3.179
600	0.0033388982	0.0024271845	2.1313706e-05	3.124	1.8170312e-05	3.040
1000	0.002002002	0.0014548982	4.6110156e-06	2.993	3.8994329e-06	3.009
SBP63						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.15	1.0046916	-	0.97105307	-
25	0.083333333	0.06	0.8728145	0.143	0.55098483	0.578
50	0.040816327	0.029411765	0.036671067	4.441	0.03166244	4.002
100	0.02020202	0.014563107	0.004657401	2.934	0.0020340242	3.903
200	0.010050251	0.0072815534	0.00027272192	4.064	0.00013119992	3.926
600	0.0033388982	0.0024271845	1.7347279e-05	2.500	2.0579042e-06	3.771
1000	0.002002002	0.0014548982	3.0936678e-06	3.371	2.7889896e-07	3.907
SBP44						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.15	1.1194001	-	0.95968249	-
25	0.083333333	0.06	0.88022901	0.245	0.67837237	0.354
50	0.040816327	0.029411765	0.06321685	3.690	0.039471219	3.985
100	0.02020202	0.014563107	0.0038336392	3.985	0.0023787988	3.994
200	0.010050251	0.0072815534	0.00023518147	3.998	0.00014602487	3.997
600	0.0033388982	0.0024271845	2.9465764e-06	3.974	1.7778227e-06	4.001
1000	0.002002002	0.0014548982	3.8215999e-07	3.993	2.2974042e-07	4.000

Figure 7: Convergence test results for the SBP-SAT scheme with $T = 1.5$

SBP21						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.15384615	1.0025843	-	0.90979831	-
25	0.083333333	0.060606061	1.467968	-0.389	1.2124351	-0.293
50	0.040816327	0.029411765	1.2301504	0.248	0.79080251	0.599
100	0.02020202	0.01459854	0.29935221	2.009	0.17951718	2.108
200	0.010050251	0.0072992701	0.071483968	2.051	0.043298928	2.037
600	0.0033388982	0.0024271845	0.0077771854	2.013	0.0047469295	2.006
1000	0.002002002	0.0014556041	0.0027835929	2.009	0.001705666	2.001
SBP42						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.15384615	1.0312329	-	0.87061186	-
25	0.083333333	0.060606061	1.388919	-0.304	0.83988649	0.037
50	0.040816327	0.029411765	0.088063332	3.864	0.053980448	3.845
100	0.02020202	0.01459854	0.0070518722	3.590	0.0047552373	3.454
200	0.010050251	0.0072992701	0.0006801628	3.350	0.00051876647	3.173
600	0.0033388982	0.0024271845	2.0238248e-05	3.190	1.8223768e-05	3.039
1000	0.002002002	0.0014556041	4.1758729e-06	3.086	3.9134295e-06	3.007
SBP63						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.15384615	1.3143241	-	1.086449	-
25	0.083333333	0.060606061	0.77383218	0.540	0.5776209	0.644
50	0.040816327	0.029411765	0.042741321	4.058	0.033193078	4.002
100	0.02020202	0.01459854	0.0060661116	2.776	0.0022122552	3.851
200	0.010050251	0.0072992701	0.00090029522	2.732	0.00015947933	3.767
600	0.0033388982	0.0024271845	2.5051445e-05	3.250	2.3482086e-06	3.828
1000	0.002002002	0.0014556041	3.5702197e-06	3.809	2.9047457e-07	4.086
SBP44						
nx	dx	dt	max_err	p_inf	l2_err	p_l2
10	0.22222222	0.15384615	1.0393717	-	0.85796278	-
25	0.083333333	0.060606061	1.1810463	-0.130	0.76590833	0.116
50	0.040816327	0.029411765	0.066700664	4.026	0.04089798	4.105
100	0.02020202	0.01459854	0.004454803	3.848	0.0024819674	3.984
200	0.010050251	0.0072992701	0.00026884546	4.021	0.00015312187	3.990
600	0.0033388982	0.0024271845	3.2377388e-06	4.010	1.8722511e-06	3.997
1000	0.002002002	0.0014556041	4.1659744e-07	4.009	2.4216632e-07	3.999

Figure 8: Convergence test results for the SBP-SAT scheme with $T = 2.0$